



## LMMSE vs MMSE

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- Suppose we wish to estimate  $\mathbf{x}$  from  $\mathbf{y}$ .

### LMMSE objective

Find  $\mathbf{A}$  and  $\mathbf{b}$  such that

$$\hat{\mathbf{x}}(\mathbf{y}) = \mathbf{A}\mathbf{y} + \mathbf{b}$$

yields the smallest possible MSE,

$$\mathbb{E}\{(\mathbf{x} - \hat{\mathbf{x}})^T(\mathbf{x} - \hat{\mathbf{x}})\}$$

.

### MMSE objective

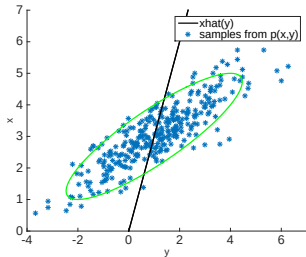
Find  $\hat{\mathbf{x}}(\mathbf{y})$  that minimizes

$$\mathbb{E}\{(\mathbf{x} - \hat{\mathbf{x}}(\mathbf{y}))^T(\mathbf{x} - \hat{\mathbf{x}}(\mathbf{y})) | \mathbf{y}\}.$$

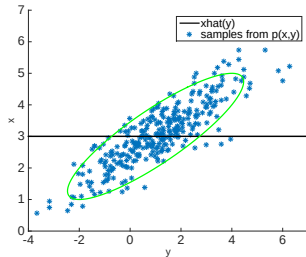
We know that the MMSE estimator is

$$\hat{\mathbf{x}}(\mathbf{y}) = \mathbb{E}\{\mathbf{x} | \mathbf{y}\}.$$

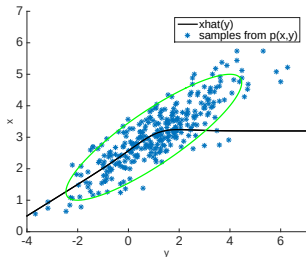
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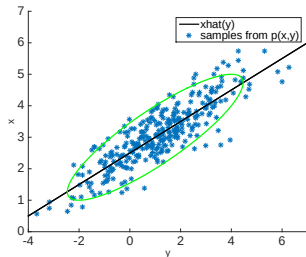
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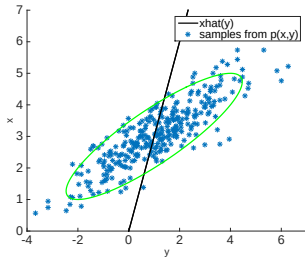
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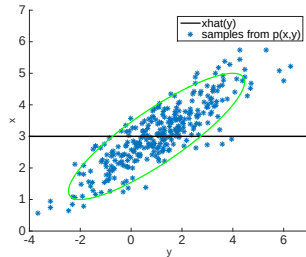
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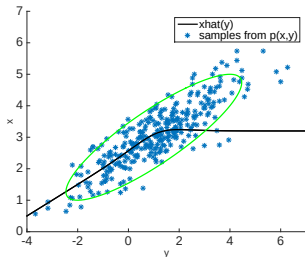
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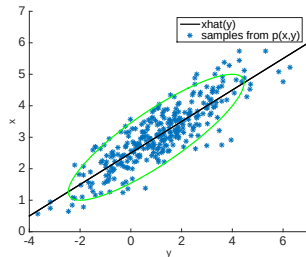
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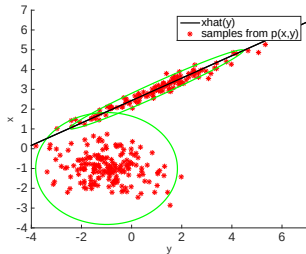
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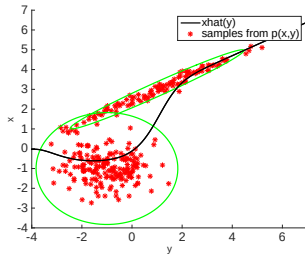
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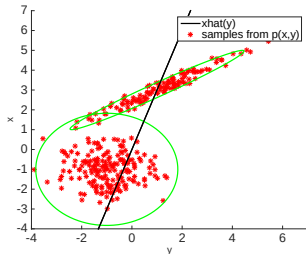
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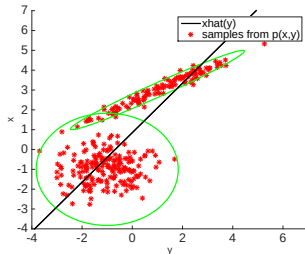
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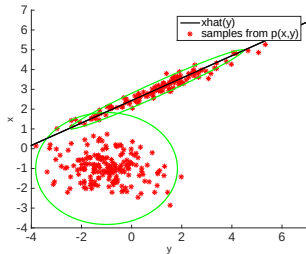
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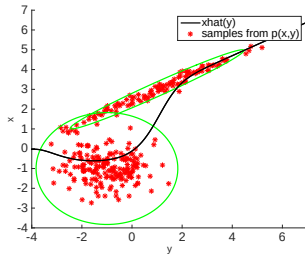
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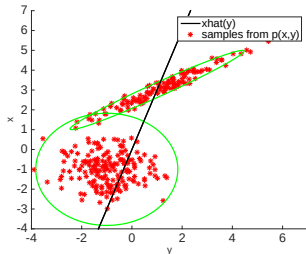
Yellow



Orange



Green



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